AI Research Methods in Finance

Spring 2027

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Course Description

This course trains students to conduct empirical finance research using artificial intelligence. Students will be taught how to employ finance theory and statistical methods to specify a well-defined research question. Students will be taught how to execute such a research inquiry using artificial intelligence tools.

Course Materials

Course materials will be provided through GitHub with an accompanying AI Assistant (see "Working with the AI Assistant" below). The course repository is available at: `https://github.com/cryptoeconprof/ai-research-methods-in-finance`. Students are responsible for keeping an updated copy of the repository and can ask the AI Assistant to download updates while preserving their work.

Students may use any tools they prefer. The recommended setup is the ChatGPT desktop app:

1. Install the ChatGPT desktop app and sign in with a ChatGPT account.
2. Select ChatGPT, switch to Work, and open a local folder for the course. Ask Work to clone the course's GitHub repository into that folder.
3. Select Codex, open the cloned repository as the course project, and set it as the project's primary folder. Start a new conversation to work with the AI Assistant.

No GitHub account or fork is required to download the repository or its updates.

Course Project

Students will complete a team research project in teams of no more than three. Each team will develop an empirical model that explains prediction-market returns. Detailed instructions—including the data scope, deliverables, deadlines, and evaluation criteria—will be announced separately.

Class Schedule and Readings

Mod 4 meets Mondays and Wednesdays from March 1 to April 23. No class meets March 15 or 17 (spring break). Additional topics for the remaining meetings will be announced.

Monday, March 1: Background Material

Codex setup and ordinary least squares (OLS). No assigned readings.

Wednesday, March 3: Risk and Return

Sharpe, William F. (1964). "Capital Asset Prices: A Theory of Market Equilibrium under Conditions of Risk." *Journal of Finance* 19(3): 425–442.

Merton, Robert C. (1973). "An Intertemporal Capital Asset Pricing Model." *Econometrica* 41(5): 867–887.

Monday, March 8: Contingent Claims Pricing

Black, Fischer, and Myron Scholes (1973). "The Pricing of Options and Corporate Liabilities." *Journal of Political Economy* 81(3): 637–654.

Cox, John C., Stephen A. Ross, and Mark Rubinstein (1979). "Option Pricing: A Simplified Approach." *Journal of Financial Economics* 7(3): 229–263.

Wednesday, March 10: Volatility Dynamics

Engle, Robert F. (1982). "Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation." *Econometrica* 50(4): 987–1007.

Bollerslev, Tim (1986). "Generalized Autoregressive Conditional Heteroskedasticity." *Journal of Econometrics* 31(3): 307–327.

Monday, March 22: Economics of Volatility

French, Kenneth R., G. William Schwert, and Robert F. Stambaugh (1987). "Expected Stock Returns and Volatility." *Journal of Financial Economics* 19(1): 3–29.

Andersen, Torben G. (1996). "Return Volatility and Trading Volume: An Information Flow Interpretation of Stochastic Volatility." *Journal of Finance* 51(1): 169–204.

Wednesday, March 24: The Relationship Between Firm Debt and Equity

Merton, Robert C. (1974). "On the Pricing of Corporate Debt: The Risk Structure of Interest Rates." *Journal of Finance* 29(2): 449–470.

Engle, Robert F., and Emil N. Siriwardane (2018). "Structural GARCH: The Volatility-Leverage Connection." *Review of Financial Studies* 31(2): 449–475.

Monday, March 29: Credit Markets

Duffie, Darrell, and Kenneth J. Singleton (1999). "Modeling Term Structures of Defaultable Bonds." *Review of Financial Studies* 12(4): 687–720.

Huang, Jing-Zhi, and Ming Huang (2012). "How Much of the Corporate-Treasury Yield Spread Is Due to Credit Risk?" *Review of Asset Pricing Studies* 2(2): 153–202.

Wednesday, March 31: Monetary Policy and Banking

Bernanke, Ben S., and Alan S. Blinder (1988). "Credit, Money, and Aggregate Demand." *American Economic Review* 78(2): 118–133.

Kashyap, Anil K., and Jeremy C. Stein (2000). "What Do a Million Observations on Banks Say about the Transmission of Monetary Policy?" *American Economic Review* 90(3): 407–428.

Drechsler, Itamar, Alexi Savov, and Philipp Schnabl (2017). "The Deposits Channel of Monetary Policy." *Quarterly Journal of Economics* 132(4): 1819–1876.

Monday, April 5: Capital Structure

Lemmon, Michael L., Michael R. Roberts, and Jaime F. Zender (2008). "Back to the Beginning: Persistence and the Cross-Section of Corporate Capital Structure." *Journal of Finance* 63(4): 1575–1608.

DeAngelo, Harry, and Richard Roll (2015). "How Stable Are Corporate Capital Structures?" *Journal of Finance* 70(1): 373–418.

Graham, John R., Mark T. Leary, and Michael R. Roberts (2015). "A Century of Capital Structure: The Leveraging of Corporate America." *Journal of Financial Economics* 118(3): 658–683.

Wednesday, April 7: Risk Pricing

Jackwerth, Jens Carsten (2000). "Recovering Risk Aversion from Option Prices and Realized Returns." *Review of Financial Studies* 13(2): 433–451.

Giesecke, Kay, Francis A. Longstaff, Stephen M. Schaefer, and Ilya A. Strebulaev (2011). "Corporate Bond Default Risk: A 150-Year Perspective." *Journal of Financial Economics* 102(2): 233–250.

Working with the AI Assistant

The AI Assistant can help with course logistics, course materials, assignments, coding, writing, literature surveys, and any other research tasks. Students remain responsible for everything they submit, including its correctness, sources, attribution, and compliance with assignment requirements. Ethical obligations, including honesty and confidentiality, still apply. Course requirements, grading decisions, and extensions remain the instructor's responsibility.